


Belief-space DP.

$$u(t) \in \mathbb{R}^{n \times p} \quad \tilde{C} = \left(C \mid \begin{matrix} \text{vec}(M^{(1)})^T \\ \vdots \\ \text{vec}(M^{(m)})^T \end{matrix} \right) \in \mathbb{R}^{m \times (n+mp)}$$

$$x(t+1) = Ax(t) + Bu(t) + w(t)$$

$$y(t) = \tilde{C} \begin{pmatrix} x(t) \\ \text{vec}(x(t)x(t)^T) \end{pmatrix} + v(t) \in \mathbb{R}^m$$

Non-gaussian

$$b(t) \propto p(x(t) | y(1:t)), \text{ no longer gaussian } \therefore p(x(t) | y(1:t)) \propto p(y(t) | x(t)) \cdot p(x(t) | y(1:t-1))$$

$$V(b(t)) = \min_{u(t)} \left[E(x(t)^T Q x(t) + u(t)^T R u(t)) + V(b(t+1)) \right] \dots (1) \quad \checkmark \text{ check.}$$

$$\begin{aligned} V(b(t+1)) &= \int x^T Q_F x \cdot p(x | y(1:t+1)) dx \\ &= E[x^T Q_F x | y(1:t+1)] = \text{Tr}(Q_F E[x x^T | y(1:t+1)]) = \text{Tr}(Q_F \widehat{x x^T}) \end{aligned}$$

By QKF.



Goal: Find $u^*(T)$ s.t. $\text{vec}(Q_F)^T \frac{\partial \{ \widehat{\text{vec}(x(T+1)x(T+1)^T)} \}}{\partial u(T)} = 0.$
 $\therefore \text{Tr}(A^T B) = \text{vec}(A)^T \text{vec}(B)$

$$\widehat{\text{vec}(x(T+1)x(T+1)^T)}^T = \text{vec}(\widehat{x(T+1)x(T+1)^T}) + K(T+1)(Y(T+1) - \widehat{Y}(T+1|T)) \quad \text{we're focusing on this term.}$$

$$= " + P_{C(T+1|T)}^T \tilde{C}^T (\tilde{C} P_{C(T+1|T)}^T \tilde{C}^T + V)^{-1} \left(\tilde{C} \left(\frac{A(x(T)) - \widehat{x(T|T)) + w(T)}{\text{vec}(x(T)x(T)^T) - \widehat{\text{vec}(x(T)x(T)^T)}} \right) + v(T+1) \right)$$

consider only n^2 part!

$$\tilde{A}(T) = \left(\begin{array}{c|c} A & 0 \\ \hline (B u(T)) \otimes A + A \otimes (B u(T)) & A \otimes A \end{array} \right) = \begin{pmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ \tilde{A}_{21} & \tilde{A}_{22} \end{pmatrix}, \quad F^u = I_n \otimes (B u(T) + A x(T)) + (B u(T) + A x(T)) \otimes I_n, \quad \tilde{C} = (C \mid M^T)$$

$$\text{vec}(\widehat{x(T+1)x(T+1)^T}) = [\tilde{A}_{21}(T) \otimes A + A \otimes \tilde{A}_{21}(T)] \widehat{x(T|T)} + (A \otimes A) \text{vec}(\widehat{x(T)x(T)^T}) + \text{vec}(\tilde{A}_{22}(T) \tilde{A}_{22}^T + w)$$

$$\widehat{\text{vec}(x(T+1)x(T+1)^T)}^T = \text{vec}(\widehat{x(T+1)x(T+1)^T}) + (N + \theta^u) (C D + C G^u + M^T N + M^T \theta^u + V)^{-1} (M^T \tilde{A}_{21}^u \tilde{x}(T) + s)$$

$$(\tilde{A}_{11} P_{11} \tilde{A}_{11}^T + w) C^T + \tilde{A}_{11} P_{12} \tilde{A}_{22}^T M^T = D \in \mathbb{R}^{n \times m}$$

$$(\tilde{A}_{11} P_{11} \tilde{A}_{21}^T + W F^u) M^T = G^u \in \mathbb{R}^{n \times m}$$

$$\tilde{A}_{22} P_{21} \tilde{A}_{11}^T + \tilde{A}_{22} P_{22} \tilde{A}_{22}^T M^T + L M^T = N \in \mathbb{R}^{n^2 \times m}$$

$$(\tilde{A}_{21}^u P_{11} \tilde{A}_{11}^T + F^u W) C^T + (\tilde{A}_{21}^u P_{11} \tilde{A}_{21}^T + \tilde{A}_{22} P_{22} \tilde{A}_{21}^T + \tilde{A}_{21}^u P_{22} \tilde{A}_{22}^T + F^u W F^u) M^T = \theta^u \in \mathbb{R}^{n^2 \times m}$$

$$C(A(x(T)) - \widehat{x(T|T)}) + w(T) + M^T \left(\tilde{A}_{22} (\text{vec}(x(T)x(T)^T) - \widehat{\text{vec}(x(T)x(T)^T)}) \right) + v(T+1) = s \in \mathbb{R}^{m \times 1}$$

$$\widehat{\text{Vec}(\mathbf{x}(t+1|\tau+1)\mathbf{x}(t+1|\tau+1)^T)} = \widehat{\text{Vec}(\mathbf{x}(t+1|\tau)\mathbf{x}(t+1|\tau)^T)} + (N+\theta^u) \left[M''(N+\theta^u) + \omega I \right]^{-1} (M''\tilde{A}_{21}^u \tilde{\mathbf{x}}(t) + S)$$

when $C=0$
 $\omega \approx 0$.

$$= \widehat{\text{Vec}(\mathbf{x}(t+1|\tau)\mathbf{x}(t+1|\tau)^T)} + (N+\theta^u) \left[(N+\theta^u)^T M''^T \right] M'' (\tilde{A}_{21}^u \tilde{\mathbf{x}}(t) + S)$$

$$= \widehat{\text{Vec}(\mathbf{x}(t+1|\tau)\mathbf{x}(t+1|\tau)^T)} + \tilde{A}_{21}^u \tilde{\mathbf{x}}(t) + S$$

$$= \tilde{A}_{21}^u \mathbf{x}(t) + (A \otimes A) \text{Vec}(\mathbf{x}(t)\mathbf{x}(t)^T) + \text{Vec}((Bu(t))(Bu(t))^T + W) + \tilde{A}_{21}^u \tilde{\mathbf{x}}(t) + S$$

$$= [(Bu(t)) \otimes A] \mathbf{x}(t) + [A \otimes (Bu(t))] \mathbf{x}(t) + \frac{\text{Vec}((Bu(t))(Bu(t))^T)}{(B \otimes B) \text{Vec}(u(t)u(t)^T)} + [(Bu(t)) \otimes A] \tilde{\mathbf{x}}(t) + [A \otimes (Bu(t))] \tilde{\mathbf{x}}(t) + \underbrace{\text{Vec}((Bu(t))(Bu(t))^T)}_{\text{independent of } u(t)}$$

$$\frac{\partial \widehat{\text{Vec}(\mathbf{x}(t+1|\tau+1)\mathbf{x}(t+1|\tau+1)^T)}}{\partial u(t)} = (B \otimes A \mathbf{x}(t)) + (A \mathbf{x}(t) \otimes B) + (B \otimes B) [u \otimes I_p + I_r \otimes u] + (B \otimes A \tilde{\mathbf{x}}(t)) + (A \tilde{\mathbf{x}}(t) \otimes B)$$

(Derivative

From Eq. (1),

$$V(b(t)) = \min_{u(t)} \left[E(\mathbf{x}(t) \otimes \mathbf{x}(t) + u(t)^T R u(t)) + \underbrace{V(b(t+1))}_{\dots} \right] \quad (1)$$

$$\text{Vec}(Q_f)^T \left[(B \otimes B) [u \otimes I_p + I_r \otimes u] + \square' \right] + \underline{2Ru(t)} = 0$$

$u \otimes I_p = T(u(t) \otimes I_p)$ where $T \in \mathbb{R}^{r \times p}$ is commutation matrix.

$$\text{Vec}(Q_f)^T \left[(B \otimes B) [u \otimes I_p + T(u \otimes I_p)] + \square' \right] + 2Ru = 0$$

$$\underbrace{\text{Vec}(Q_f)^T}_{q^T} \left[\underbrace{(B \otimes B)}_{n^2 \times p^2} \left(\underbrace{I_p + T}_{p^2 \times p^2} \right) \underbrace{(u \otimes I_p)}_{p^2 \times p} + \underbrace{\square'}_{n^2 \times p} \right] + \underbrace{2Ru}_{p} = 0$$

$\square' = (B \otimes A \mathbf{x}(t)) + (A \mathbf{x}(t) \otimes B) + (B \otimes A \tilde{\mathbf{x}}(t)) + (A \tilde{\mathbf{x}}(t) \otimes B)$

Transpose & rearrange

$$(A(u) + \square')^T q + 2Ru = 0$$

Once we define $S = \sum_{i=1}^r [M(e_i \otimes I_p)]^T q e_i^T$

$$(A(u) + \square')^T q = Su + \square'^T q$$

$u \otimes I_p = \sum_{i=1}^p u_i (e_i \otimes I_p) \in \mathbb{R}^{r \times r}$
 $M(u \otimes I_p) = \sum_{i=1}^p u_i M(e_i \otimes I_p)$ } Linearity.

Therefore $Su + 2Ru + \square'^T q = 0 \quad \therefore u = -(S + 2R)^{-1} \square'^T \text{Vec}(Q_f)$ ~~is the~~ analytical solution.